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Autonomous Institute
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DTE Code : 3218 AICTE Permanent ID : 1-4790201
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Synopsis

Smart Inventory & Warehouse Management System

— Hardware Module (RFID, Load Cell & ESP32 Subsystem)

Submitted in partial fulfilment of the requirements of the degree of

Bachelor of Engineering

In

Computer Science and Engineering (Data Science) by

Princy Patil [Roll No. 37, PID No. EU1258010]

Prinkal Bari [Roll No. 03, PID No. EU1258024]

Siddhi Raut [Roll No. 40, PID No. EU1258018]

Snehal More [Roll No. 25, PID No. EU1258020]

Under the Guidance of

(Prof. Sachin Sase)
Department, SJCEM

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI

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Certificate of Approval

This is to certify that Princy Patil (Roll No. 37), Prinkal Bari (Roll No. 03), Siddhi Raut (Roll No. 40) and Snehal More (Roll No. 25), are Bonafide students of B. Tech in Computer Science and Engineering (Data Science)- Department, St. John College of Engineering and Management, Mumbai. They have satisfactorily completed the requirements of Synopsis as prescribed by **ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT (An Autonomous College affiliated to University of Mumbai),** while working on "Smart Inventory & Warehouse Management System".

Signature: ----------
Name: Prof. Sachin Sase
Designation: Assistant Professor

Signature : ----------
Name: Joslyn Gracias, Head of Department
Designation: HOD of CSE(DS)

Signature:
Name: Dr. Kamal Shah
Principal,
St. John College of Engineering and Management

Declaration

I/we declare that this written submission represents my/our ideas in my/our own words and where others ideas or words have been included, I/we have adequately cited and referenced the original sources. I/we also declare that I/we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my/our submission. I/we understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

(Signature)

Princy Patil (Roll No. 37, PID No. EU1258010)
Prinkal Bari (Roll No. 03, PID No. EU1258024)
Siddhi Raut (Roll No. 40, PID No. EU1258018)
Snehal More (Roll No. 25, PID No. EU1258020)

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(Student Name)

Date:

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Group Members

1. Princy Patil (Roll No. 37, PID No. EU1258010)
2. Prinkal Bari (Roll No. 03, PID No. EU1258024)
3. Siddhi Raut (Roll No. 40, PID No. EU1258018)
4. Snehal More (Roll No. 25, PID No. EU1258020)

Abstract

This report covers the hardware subsystem of the Smart Inventory & Warehouse Management System: an IoT-enabled identification and sensing layer built around RFID and load-cell technology. RFID is used for unique product identification and stock-in / stock-out movement tracking.

A load cell with an HX711 amplifier is used for weight-based stock estimation, particularly for unpacked or quantity-by-weight goods. An ESP32 controller acquires these hardware readings and communicates them over Wi-Fi to the application layer.

This report focuses specifically on the physical sensing and communication layer — RFID, load cell/HX711 and ESP32 — and does not cover the camera-based OCR label-reading pipeline, which is addressed separately in the corresponding software module report under the project's software guides.

At the current stage, the hardware architecture and data-flow have been designed; component selection, prototyping, calibration and ESP32 firmware development are positioned as the subsequent implementation and validation work.

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Abbreviations and Symbols

- IoT: Internet of Things
- RFID: Radio Frequency Identification
- HX711: Load-cell amplifier/interface IC
- Wi-Fi: Wireless Fidelity (wireless networking)
- API: Application Programming Interface
- WBS: Work Breakdown Structure
- SDG: Sustainable Development Goal
- BOM: Bill of Materials

Chapter 1: Introduction

1.1 Background

Inventory systems depend on reliable, low-effort identification of physical goods and accurate measurement of quantity on hand. This report addresses the hardware identification and sensing layer of the Smart Inventory & Warehouse Management System: RFID for unique product identification and movement tracking, and load-cell + HX711 sensing for weight-based stock estimation, both coordinated through an ESP32 controller that communicates over Wi-Fi to the software backend.

This hardware layer is designed to demonstrate practical application of embedded systems, IoT communication and sensor interfacing, complementing the OCR-based label-reading and software decision-support work carried out under the project's software module.

1.2 Problem Statement

Manual stock counting and tagging-free tracking make it difficult to know, at any given moment, which physical items are present and how much stock is actually on hand. A unique, machine-readable identity (RFID) is needed for tagged items, and a non-manual weight-based measurement path is needed for unpacked or bulk goods where individual tagging is impractical.

1.3 Need of the Project

A dependable hardware sensing and communication layer is required so that physical stock movement and quantity can be captured automatically and transmitted reliably to the software system, instead of relying on manual counts and entries.

1.4 Objectives

- Design an IoT-enabled hardware layer that links physical warehouse inputs to the software inventory system.
- Use RFID for unique product identification and stock-in / stock-out movement tracking.
- Use load-cell measurement with HX711 to estimate stock quantity for weight-based products.

- Use ESP32 for hardware input collection, local processing and Wi-Fi communication with the backend.
- Ensure reliable, low-duplication transmission of hardware readings to the application layer.

1.5 Scope

Covered: RFID-based identification and movement events, load-cell/HX711 weight sensing, ESP32 data acquisition and Wi-Fi transmission.

Out of scope for this report: camera-based OCR label reading, backend/database development, dashboard and business-logic/decision rules — these are covered in the corresponding software module report.

Limitations: component selection, prototyping, calibration and firmware development are at the design/planned stage and not yet completed.

1.6 Expected Outcomes

- A working hardware prototype demonstrating RFID-based identification and movement tracking.
- A calibrated load-cell + HX711 weight-measurement path.
- ESP32 firmware capable of acquiring sensor/RFID readings and transmitting them reliably over Wi-Fi.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews existing work related to RFID-based identification and IoT weightsensing systems, in order to position the proposed hardware layer relative to prior approaches.

2.2 Literature Review Table

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Paper	Author	Year	Method	Findings	Research Gap
ESP32 Series Datasheet	Espressif Systems	—	Technical documentation	Confirms Wi-Fi/Bluetooth connectivity and peripheral set suitable for connected prototyping	N/A — reference standard
RFID Standards Overview	GS1	—	Standards documentation	Describes EPC identifiers and RFID use in supply-chain contexts	Does not itself address weightsensing fusion
[To be added by team]					

2.3 Research Gap

- Prior hardware-focused work typically addresses either RFID identification or weight-based sensing, rather than combining both through a single low-cost controller.

- This module addresses the gap by using RFID for identity/movement and load-cell sensing for weight-based quantity, both acquired and transmitted through a common ESP32 layer.

2.4 Novelty / Innovation

- A combined RFID + load-cell sensing path through a single ESP32 controller, each mechanism assigned a distinct, non-overlapping role.
- Design oriented toward low-duplication, reliable event transmission over Wi-Fi to the backend.

Chapter 3: Proposed Methodology

3.1 Proposed System

The proposed hardware module uses RFID for unique identification and movement tracking, and a load cell + HX711 for weight-based stock estimation, especially for unpacked or quantity-by-weight goods. An ESP32 acquires both readings and transmits them over Wi-Fi to the backend API maintained by the software module.

3.2 Working Principle

- A tagged product or weighable item enters the warehouse workflow.
- RFID identifies the tagged physical item through its unique identifier; RFID events represent stock-in and stock-out movement.
- The load cell measures physical weight, with HX711 as the interface/amplifier to the ESP32, useful for unpacked or weight-based products.
- ESP32 collects the RFID/weight readings, performs local processing and prepares the data for transmission over Wi-Fi.
- The ESP32 transmits the readings to the backend API over Wi-Fi for further processing by the software module.

3.3 System Architecture / Block Diagram

The hardware module is organized into three logical layers: physical sensing, the ESP32 controller, and the communication layer to the backend.

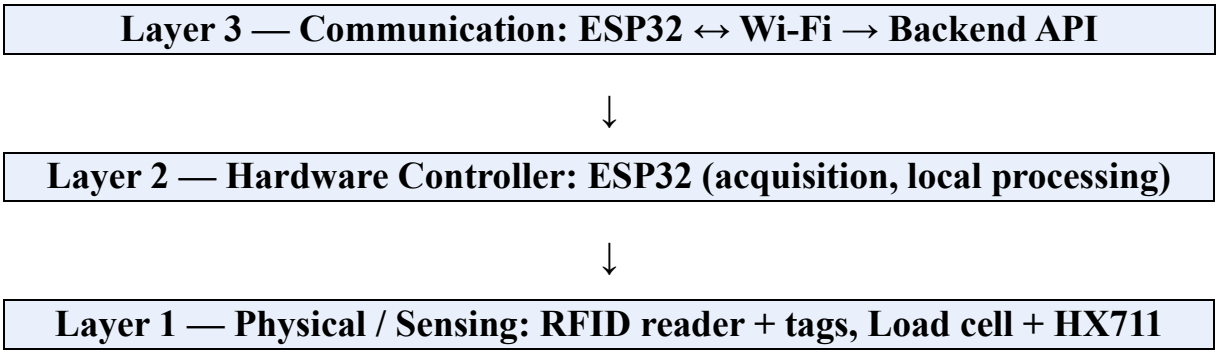


Figure 2. Hardware-layer block diagram.

3.4 Flow Chart

Proposed hardware data flow, from product entry to transmission into the software backend:

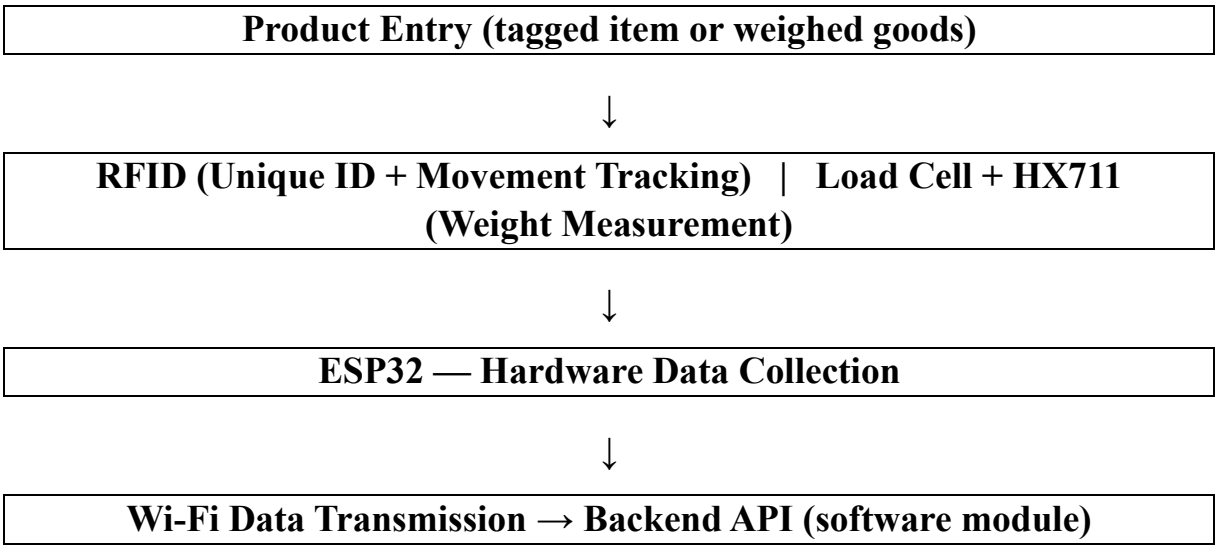


Figure 1. Proposed data flow for this module (target design, not a claim of completed integration).

3.5 Methodology

Design procedure:

Requirement analysis → hardware architecture design → RFID and load-cell component selection → ESP32 firmware design → communication protocol design (Wi-Fi → backend API).

Experimental / development procedure:

Hardware preparation and load-cell calibration → RFID reader/tag testing → ESP32 firmware development → Wi-Fi transmission testing → module-level hardware testing before handover to the software backend.

Programming steps: ESP32 firmware for RFID/load-cell acquisition and Wi-Fi transmission to the backend API.

Prototype setup and arrangement:

The photographs below show the current physical prototype arrangement prepared for the Smart Inventory & Warehouse Management System. Sample goods are arranged in separate wooden storage sections, with the RFID reader module, ESP32 controller and connecting wires positioned in front of the rack. This arrangement illustrates the planned hardware layout for product identification and system integration. The photographs document the prototype setup; they do not, by themselves, confirm completed load-cell calibration or end-to-end backend communication.



Figure 3. Close-up view of the prototype hardware and storage sections.



Figure 4. Overall view of the prototype rack and controller arrangement.

3.6 Components / Materials

Component	Specification / Role	Quantity
ESP32	Main embedded controller; Wi-Fi communication and sensor/RFID input handling	1 (per node)
RFID Reader	Reads RFID tag identifiers for product identification and movement events	TBD
RFID Tags	Carry unique identifiers on tagged products	TBD
Load Cell	Measures physical weight on a weighing platform	1 per weighing station
HX711	Amplifier/interface between load cell and ESP32	1 per load cell
Display / LED / Buzzer (optional)	Local feedback or notification	Optional — not confirmed

Final specifications and quantities are to be confirmed during implementation (Chapter 4).

Chapter 4: Project Planning

4.1 Work Breakdown Structure

- 1. Requirement analysis and hardware scope definition
- 2. Hardware architecture design
- 3. RFID reader/tag and load-cell/HX711 component selection
- 4. Hardware prototyping and sensor/reader validation
- 5. Load-cell calibration
- 6. ESP32 firmware development (acquisition + Wi-Fi transmission)
- 7. Hardware-to-backend communication testing
- 8. Hardware testing and debugging
- 9. Handover / integration support to the software module

4.2 Gantt Chart

Semester planning — presented phase-wise since fixed calendar dates were not finalized at the time of writing.

Phase	Focus	Status
Phase 1	Requirements + hardware scope	Completed / defined
Phase 2	Hardware architecture + data flow	Completed / designed
Phase 3	Component selection (RFID, load cell, HX711, ESP32)	Planned
Phase 4	Hardware prototyping + calibration	Planned
Phase 5	ESP32 firmware development	Planned
Phase 6	Wi-Fi → backend communication testing	Planned
Phase 7	Hardware testing + debugging	Planned
Phase 8	Integration support to software module	Planned

4.3 Roles and Responsibilities

Student	Responsibility
Student 1	RFID reader/tag integration and testing
Student 2	Load cell + HX711 calibration
Student 3	ESP32 firmware development
Student 4	Hardware-to-backend communication testing

4.4 Risk Assessment

Risk	Impact	Mitigation
Load-cell calibration / unstable weight readings	Incorrect quantity estimation	Careful calibration procedure and repeated validation before deployment
RFID reading reliability / tagreader placement	Missed or duplicate movement events	Controlled placement testing and duplicate-event detection logic
Hardware–software communication / Wi-Fi reconnection	Data loss or delayed updates	Retry/reconnection handling in ESP32 firmware
Component availability / lead time	Delayed prototyping	Early finalization of component list and vendor sourcing

Chapter 5: Resources & Feasibility

5.1 Technical Feasibility

The hardware module uses commonly available embedded and sensing technologies: ESP32 for connected control, RFID for identification, and load-cell/HX711 sensing. Espressif's official documentation confirms Wi-Fi capability and a broad peripheral set across the ESP32 family, supporting the suitability of the controller for this prototype.

5.2 Economic Feasibility

The hardware design relies on relatively accessible development components (ESP32, RFID reader/tags, load cell, HX711). A detailed bill of materials and total cost have not yet been finalized (see Section 5.4).

5.3 Operational Feasibility

The hardware layer is designed to operate with minimal manual intervention once calibrated, automatically forwarding identification and weight readings to the backend.

5.4 Budget Estimation

Table 5.1 lists the components and prices transcribed from the two New Silikon Electronics estimates. Estimate No. 320 totals ₹2,380 and Estimate No. 321 totals ₹70, giving a combined estimated cost of ₹2,450.

Table 5.1: Itemized Hardware Resources and Budget Estimation

No.	Name of Component	Quantity	Unit Cost (₹)	Total Cost (₹)
1	ESP32 development board	1	400	400
2	RC522 RFID reader module	1	100	100
3	RFID keychain tags	6	20	120
4	10 kg load cell	1	150	150
5	HX711 load-cell amplifier module	1	45	45
6	Jumper wires (as written on estimate)	3	50	150
7	MB102 breadboard power module	2	65	130
8	5 V / 1 A power adapter	1	110	110
9	FT232 USB-to-serial adapter	1	100	100
10	Resistors	10	1	10
11	Breadboard	1	90	90

No.	Name of Component	Quantity	Unit Cost (₹)	Total Cost (₹)
12	USB mini cable	2	35	70
	Estimate No. 320 subtotal	—	—	1,405
	Estimate No. 321 subtotal	—	—	70
	TOTAL ESTIMATED COST	—	—	1,475

Note: Unit rates and amounts are transcribed from the supplied handwritten estimates. The description of the ₹750 item is unclear in the estimate and must be verified with the vendor before final submission or purchase.

5.5 Bill of Materials (BOM)

The itemized bill of materials, quantities, unit rates and estimated amounts are presented in Table 5.1.

- ESP32 development board
- RFID reader module and RFID tags
- Load cell (suitable capacity) + HX711 amplifier
- Optional: display / LED / buzzer for local feedback

Chapter 6: Expected Results & Impact

6.1 Expected Outcomes

- Reliable RFID-based identification and stock movement events.
- Calibrated, accurate weight-based quantity estimation.

6.2 Industrial Relevance

Reliable, low-cost identification and weight-sensing hardware is broadly applicable across warehouses, retail stores and small-to-medium supply-chain operations.

6.3 Societal Impact

By supporting more timely and accurate inventory records, the module can help reduce stockouts and overstocking, contributing to more reliable availability of goods for end consumers.

6.4 Environmental Impact

Accurate hardware-based stock measurement reduces the need for repeated manual counting activity.

6.5 SDG Mapping

- SDG 9 — Industry, Innovation and Infrastructure
- SDG 12 — Responsible Consumption and Production
- SDG 8 — Decent Work and Economic Growth

6.6 Future Scope

- Additional sensor types (e.g. temperature/humidity) on the same ESP32 hardware layer.
- Multiple RFID/weighing stations across a multi-location warehouse.
- Battery-powered or low-power hardware nodes for mobile use.

Chapter 7: Conclusion

This report presents the hardware subsystem of the Smart Inventory & Warehouse Management System: RFID-based identification and movement tracking, load-cell + HX711 weight sensing, and an ESP32 controller that acquires and transmits this data over Wi-Fi to the backend.

By assigning distinct roles to RFID and weight sensing, the module provides a reliable low-cost identification and measurement layer for the overall system.

At this stage, the hardware architecture and data flow have been designed; component selection, prototyping, calibration and firmware development remain as the primary next steps.

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Appendices (Optional)

- **Appendix A:** Hardware components photograph.
- **Appendix B:** Proposed warehouse model photograph.
- **Appendix C:** Final circuit diagram and ESP32 pin connections.
- **Appendix D:** Load-cell calibration calculations and hardware testing records.